

The Fundamentals of Programming

Modular Thinking & Libraries



Objectives

What you will learn:

1. How to use Python libraries.
2. The difference between a library and a module.
3. How to create your own module and why you would want to.

Libraries

Adding abilities

Python has a small set of functions that you can use without any extra effort, like **int**, **float**, and **min**. Other functions require you to use modules from the **Python Standard Library**. To use the modules from these libraries you need to **import** them.

```
import random  
import math  
import webbrowser
```

```
print(random.randint(0, 5))  
print(math.sqrt(64))
```

```
address = "http://www.google.com"  
webbrowser.open_new(address)
```

Import

Bring things in

The **import** keyword can be used to add these modules for use in your program, but where do you find them? You can use the documentation or search.

Do a search for the Python modules on the right and find an example of what they allow you to do.

datetime
tkinter
turtle
os
statistics

The Making of a Module

There is nothing special about modules, and it's even quite easy to make your own since they are in fact just separate Python files.

main.py

```
import module
```

```
var = module.add(5, 10)  
print(var)
```

module.py:

```
def add(a, b):  
    return a + b
```

How it works

The dot

When you import a module it makes all the functions and variables of that module available to you, but to use them you have to use the module name and the **dot** operator.

module.py:

```
def add(a, b):  
    return a + b
```

```
var = "Hi"
```

main.py:

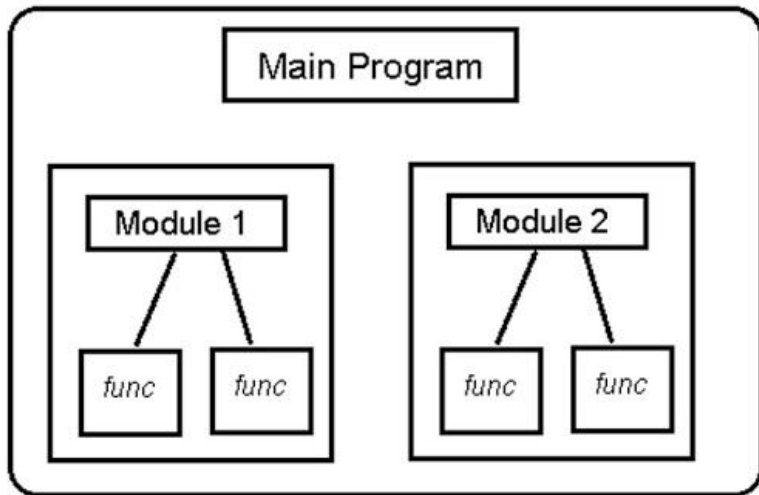
```
import module
```

```
print(module.var)  
a = module.add(5, 1) + module.add(1, 7)  
print(a)
```

Why?

Modularization

You might be wondering why you need to put things into modules. There are many reasons, but grouping similar functions, and keeping source files small are two.



Practical

Badmath

Create a module called **badmath.py** that has three functions called **add**, **multiply**, and **divide**.

Each of these functions should take two parameters **a**, and **b**. The return value should be the value **a** and a random integer.

badmath.py

```
import random
```

```
def add(a, b):
```

```
    ...
```

```
def multiply(a, b):
```

```
    ...
```


Recap

We learned

1. That a library is made of modules.
2. The Python Standard Library has many modules that you can use in your own program.
3. To create your own module all you need is to create another Python file.